Draft version September 1, 2026
Typeset using LaTeX **default** style in AASTeX7.0.1

Don’t Even Try: Auditing the Recoverability Frontier of Galaxy Deblending

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Submitted to Journal to be determined

ABSTRACT

When two galaxies overlap in an image, a deblender must decide how much of the shared light belongs to each one. It can return a realistic galaxy and still divide that light incorrectly. This report follows one such failure from reconstruction to recoverability.

A 1,927,075-parameter U-Net learned a signed blend-to-target correction and reduced affected-region mean squared error by 28.81× relative to the unchanged blend on group-disjoint normal development scenes. A separate coordinate-conditioned U-Net selected the requested source in 98.0% of prompt swaps, with 0.2% output collapse. Those results establish useful prediction and working request control, but not that the returned light belongs to the requested galaxy.

A loss-geometry audit explains why. The architecture could produce useful reconstructions, but the loss often preferred a blended compromise over the exact source pair: the exact truth minimized the training objective on only 10 of 64 rows, while a compromise scored lower on 54 of 64. An ambiguity atlas then produced observation-equivalent alternatives in 50 of 50 constructed cases, 19 of them inside the same model family. The unrestricted additive two-source map has 10,800 independent transfer directions, although nonnegativity bounds their extent, and dimension is not a photometric-severity measure. Supplying the correct shape family with answer-key source brightness produced strict uniqueness in seven of eight selected scenes; removing the hidden brightness produced none. A second exposure improved local diagnostics without restoring strict uniqueness, and an external brightness measurement helped four selected scenes. The result is an evidence policy: report a source when the stated observation and assumptions support it, seek an independent measurement when one can distinguish the alternatives, and otherwise abstain.

Note: 250-word limit for abstract

Keywords: Astronomy data analysis (1858) — Convolutional neural networks (1938) — Galaxies (573) — Astronomical methods (1043)

1. INTRODUCTION

In a deep sky image, galaxies are not neatly separated. Their light overlaps, and a deblender has to decide which photons belong to which source. That decision is not cosmetic: the brightness, the shape, and the small gravitational distortion measured for each galaxy all depend on how the shared light was divided, so an incorrect split propagates directly into photometry, morphology, and weak-lensing measurements (P. Melchior et al. 2021). As surveys grow deeper, more detected objects overlap with a neighbour, which makes the reliability of that decision a practical problem rather than an edge case.

This research project began with a machine-learning question: can a coordinate-prompted neural network reconstruct the requested galaxy from a blend? Answering it required a controlled blend simulator, a U-Net trained to undo the

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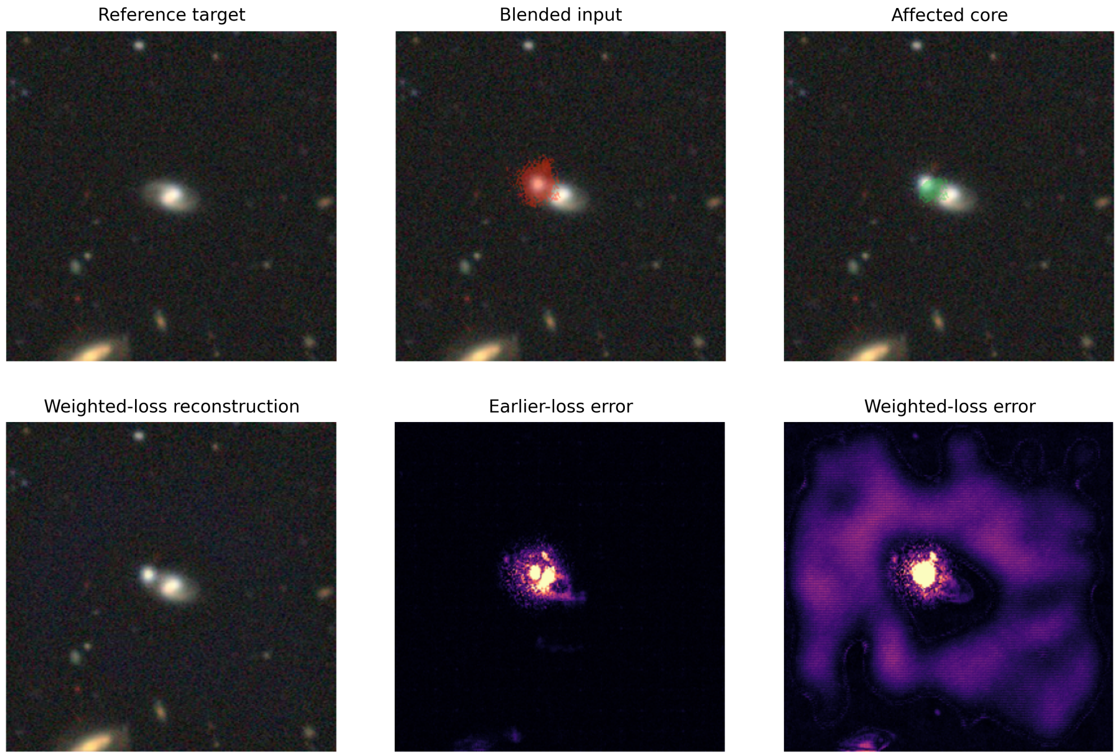

Figure 1. A believable historical failure. The weighted-loss reconstruction remains galaxy-like while retaining a second component and broad structured error. Synthetic truth exposes the mistake; an unlabeled survey blend would not provide the reference panel. The example comes from the superseded random-row development protocol and illustrates a mechanism, not grouped performance.

blending, and a test of whether a coordinate could ask the network for one specific source. Each step worked. The project nevertheless ended with a harder question: under what conditions does the available evidence justify the reconstruction the network returns?

The reason for that shift is visible in Figure 1. The failure was not an obviously broken image. It was a believable one. The reconstruction still looks like a galaxy, but it keeps part of the contaminating source and spreads structured error across the frame. The mistake is detectable here only because the scene is synthetic and the true target sits beside it as a reference panel; an unlabeled survey blend supplies no such panel, so the same error would pass an eye check. It is important to note that this example comes from the superseded random-row protocol, so it illustrates a mechanism rather than the grouped performance reported later.

This distinction matters enough to name. A *prediction* is the image a model returns. *Authorization* is the stronger statement that the stated observation, with the assumptions declared alongside it, rules out scientifically different answers. A model can be well trained, responsive to the requested coordinate, and confidently wrong about the allocation, because nothing in training asks whether a competing split of the light would have explained the same observation equally well.

The architecture campaign improved the prediction rule in four deliberate steps: it replaced a direct image target with a signed correction, sampled difficult overlaps on purpose, weighted the overlapping region inside the loss, and added a coordinate request. Each step was tested separately and each helped. However, none addressed whether the observation permitted only the returned answer.

This report therefore makes three contributions. First, we establish what the implemented U-Nets could achieve under leakage-aware development evaluation, so the positive results are stated with their evidence boundaries attached. Second, we test where the remaining failures came from, asking whether the network could not represent the correct answer, whether the training objective scored it poorly, whether optimization failed to reach it, or whether the observation never contained enough information to single it out. Third, we audit which assumptions and which

Table 1. Evidence summary. Each row answers a different question; no single positive result substitutes for the others.

Claim	Test	Observed result	Interpretation
Controlled fidelity	Group-disjoint reconstruction	28.81× affected-region ratio	Learned correction on synthetic RGB
Prompt response	Move only the coordinate	98.0% success; 0.2% collapse	Request controls output identity
Objective misspecification	Loss-geometry audit	Exact truth objective-optimal on 10 of 64 rows	Training loss prefers a compromise over the true decomposition
Alternative existence	Ambiguity atlas	50/50 equivalent pairs; 19/50 in-family	Plausible alternatives survive
Strong information	Shape + answer-key brightness	7/8 strict unique	Sufficient constraints can identify a source
Weaker information	Shape without hidden brightness	0/8 strict unique	Tested shape rules did not suffice
Paired exposure	S1/S2/P2 control	3 PSF, 2 exposure, 11 unassigned; 0/8 strict	Local gain is not uniqueness
Post-hoc calibration	Safe-output labeling	0/7,591 safe	No positive class for a safety classifier

Note. The 28.81 ratio uses a truth-defined affected mask and an identity baseline. The eight-scene counts describe selected discovery scenes, not population rates.

additional measurements reduce the surviving ambiguity, using constructed alternatives, multi-start fits, stronger shape priors, paired observations, and external photometry. Table 1 states each claim once, with the test that produced it and the interpretation it supports.

2. BACKGROUND

Deblending is an inverse problem: the observation is given and the sources that produced it must be inferred. A single pixel value combines source morphology, color, and position with the blurring the telescope and atmosphere apply, with noise, and with detector response. That blurring is described by the point-spread function (PSF), the image a single point of light produces after passing through the optical system. Because many source combinations survive that chain and still reproduce the same pixel values, the inverse direction is not determined by the observation alone.

Classical methods therefore add assumptions explicitly. SExtractor segments connected regions of light and assigns them to detections (E. Bertin & S. Arnouts 1996). More constrained methods declare a stronger source model before fitting: SCARLET, for example, factorizes a multiband scene into nonnegative spectral and morphological components, so every candidate source must have a nonnegative image and a consistent color (P. Melchior et al. 2018). Such declarations shrink the set of source combinations the method will consider, and they do so where a reader can inspect them.

Learned deblenders move part of that source prior somewhere less visible: it now lives in the training distribution, the architecture, and the loss (A. Boucaud et al. 2020). A U-Net suits this task because its encoder progressively summarizes the scene while its skip connections return fine spatial structure to the decoder (O. Ronneberger et al. 2015). Residual learning simplifies the task further by predicting a correction to the input rather than an entire image (K. He et al. 2016). Neither design, however, changes how many independent measurements the telescope actually made.

A coordinate addresses a different problem again. It does not describe the source; it identifies which of several sources the user is asking about. Coordinate-conditioned deblenders use that signal to extract one requested component from a crowded scene (R. Zhang et al. 2025). The coordinate says where to look, but it does not label the overlapping photons and it does not supply the brightness of the source at that position. In contrast to a nonnegativity or color constraint, it adds no restriction on how the shared light may be divided.

The gap this report addresses therefore sits after detection, reconstruction, and prompting. These tools can select a plausible source, but they do not automatically show that the observation permits only that source.

Table 2. Data representations and evidence roles. Their numerical metrics are not pooled.

Data	**Representation**	**Partitions**	**Role**
RGB sources	$256 \times 256 \times 3$ display RGB; $[0, 1]$	12,417 / 2,660 / 2,659	Architecture progression
RGB blends	Clipped simulator output	8,000 train; 1,000 val.; $4 \times 1,000$ dev.	Normal and stress fidelity
Multiband scenes	$3 \times 60 \times 60$ $g/r/z$; electrons	8,000 / 1,000 / 1,000 / 1,000	Coordinate and allocation tests
Lockbox	Same representation	Sealed / unused	No performance claim

Note. RGB source partitions are train/validation/development. Multiband partitions are train/validation/calibration/development. Counts are source rows, blends, or scenes as identified.

3. DATA AND EVIDENCE BOUNDARY

3.1. *Two noninterchangeable experimental tracks*

The project uses two data representations, and keeping them separate is part of the evidence boundary. Reconstruction experiments use 256×256 display-RGB Galaxy10 DECaLS cutouts (W. H. Leung & J. Bovy 2024; A. Dey et al. 2019); coordinate and recoverability experiments use $3 \times 60 \times 60$ detected-electron $g/r/z$ scenes. Their units, simulators, and metrics differ, so their numerical results are never pooled. The RGB track establishes controlled restoration behavior. The multiband track then asks whether the observation and declared source model identify one allocation.

3.2. *Display-RGB reconstruction track*

The RGB sources are split so no galaxy can appear on both sides of the split. The Galaxy10 file contains 17,736 unsigned 8-bit RGB rows, which the loader converts to floating point in $[0, 1]$. It is important to note that these are rendered display composites rather than calibrated linear flux, so any metric computed on them describes controlled image restoration and not survey photometry. The corrected split groups the union of exact image hashes and exact finite sky coordinates, then assigns 12,417/2,660/2,659 rows to train, validation, and development, with no group crossing a partition. It replaced an earlier row-index split containing 29 exact-pixel and 27 exact-coordinate crossing pairs, which had allowed the same source to be seen during both training and evaluation.

Each synthetic example is built from two real galaxies. The generator picks a target Y and a contaminant C, extracts and transforms the contaminant foreground, and adds it to a slightly blurred, noisy copy of the target before clipping the result back into display range:

$$X = \Pi_{[0,1]} \left[G_\sigma(Y) + S_{d_x,d_y} R_\phi \{ \alpha F(C) \} + \epsilon \right] . \tag{1}$$

Here F extracts the contaminant foreground, R_ϕ rotates it and S_{d_x,d_y} translates it without wrapping around the frame, α scales its brightness, G_σ blurs the target, ϵ adds noise, and $\Pi_{[0,1]}$ clips the sum into display range. One consequence of that final clip is that the observation is not a clean sum of two images, because bright overlaps lose information at the ceiling. At inference the model receives only X; the synthetic truth Y constructs the training target and the evaluation masks, and is never an input.

Difficult overlaps are rare when the transformation parameters are sampled uniformly, so the project sampled them deliberately. Normal blends use shifts up to 56 pixels, contaminant brightness 0.5–1.2, target blur 0–0.3 pixels, and noise standard deviation 0–0.01. Three development suites add close and brighter overlaps, compact-bright contaminants, and high-core obstruction, the regimes in which the target's own center is most strongly corrupted. Balanced training samples 50% normal cases, 30% high-overlap or core-obstructed cases, and 20% brightness/size stress cases. Frozen manifests record source groups, transformations, seeds, masks, and array hashes; all 13,000 rows replayed exactly.

3.3. *Truth-defined reconstruction metric*

Reporting error over the whole image would be misleading, because most pixels in a 256×256 cutout are background the blend never touched, and a model that changed nothing would score well by leaving them alone. The project therefore restricts the metric to pixels where the blend and the target materially disagree:

$$A(p) = \mathbb{1} \left[\frac{1}{3} \sum_{c=1}^{3} |X_c(p) - Y_c(p)| > 0.02 \right] . \tag{2}$$

The indicator $\mathbb{1}[\cdot]$ is one where the bracketed condition holds and zero otherwise, p indexes pixels, c indexes the three color channels, and the threshold 0.02 is in the same $[0,1]$ display units as the images. The mask therefore marks where the contaminant left a visible trace, before any prediction is made. What follows from that definition must be stated explicitly. Because A is built from the synthetic truth Y, it selects exactly the pixels where the unchanged blend is already wrong. That makes it a reasonable measure of how well a model repairs known affected pixels, and an unsuitable neutral region for comparing the identity baseline against a different scientific estimator.

3.4. *Multiband recoverability track*

The second, multiband track is where the information questions are asked, and it comes from a different source. Its scenes are not real cutouts but renderings of a parametric galaxy catalog, which is what makes exact isolated truth available. The catalog is the CatSim sample distributed with the Blending ToolKit (BTK) v1.0.9, whose 86,273 entries describe each galaxy by bulge, disk, and AGN flux fractions, half-light axes, position angles, and $g/r/z$ magnitudes; BTK renders those parameters through GalSim under an LSST survey model (I. Mendoza et al. 2025). It contains 8,000/1,000/1,000/1,000 train/validation/calibration/development scenes with two sources, three bands, 0.2 arcsec pixels, and nominal PSF FWHM 0.86/0.81/0.77 arcsec in $g/r/z$. Band scales come from training-only 99.5th-percentile statistics and preserve negative values without clipping, so noise fluctuating below the sky level is not discarded. Prompted and unprompted randomized models share byte-identical scenes and source requests, so any difference between them can only come from the prompt. Isolated sources are evaluation truth. Exact source brightness and truth-derived morphology enter only the labeled answer-key upper-bound audit of Section 6. The lockbox remained sealed.

3.5. *Observable information and hidden truth*

The distinction between evaluation truth and inference-time information is essential for interpreting the recoverability audit. The coordinate model observes only the normalized blend and its Gaussian coordinate plane. The structural audit additionally uses declared source families, known coordinates and PSFs, multi-start fit outputs, and any explicitly acquired external measurement. Isolated source images, catalog parameters, truth-defined masks, and answer-key source fluxes remain hidden. They may score an experiment or define an upper bound, but they cannot justify an inference-time claim. Consequently, the current truth-based reconstruction thresholds validate candidate outputs only on simulations; a future operational decision rule would require an observable, independently calibrated bound on the corresponding scientific errors.

4. RECONSTRUCTION: WHAT THE ARCHITECTURE ACHIEVED

4.1. *One fixed U-Net and four controlled changes*

The RGB campaign changed only one thing at a time. The topology was fixed at the start and never altered, so any improvement is attributable to the intervention rather than to added capacity.

The fixed network is a three-stage U-Net. Its encoder returns 32, 64, and 128 channels while three max-pooling steps halve the grid at each stage, reaching a $256 \times 32 \times 32$ bottleneck. Transposed convolutions restore the grid, and skip connections concatenate matching encoder features into decoder stages with 128, 64, and 32 output channels, so localized structure lost to pooling is returned to the output. Each block repeats a bias-free 3×3 convolution, BatchNorm, and ReLU twice, and a final 1×1 projection produces three channels. The network contains 1,927,075 trainable parameters and no attention, recurrence, dropout, or fully connected layer. Figure 2 gives the exact tensor shape at every stage.

The first model asked the most direct question available: can a network output the isolated target given only the blend? It predicted the target image itself,

$$\widehat{Y} = f_\theta(X), \tag{3}$$

where f_θ is the U-Net with parameters θ, X is the blend of Equation 1, and $\widehat{Y}$ is the prediction, held in display range by a sigmoid head. The mapping is well posed on synthetic data, where every blend has one recorded target. Its limitation is that the network must reproduce the whole frame, so much of its capacity is spent relearning background pixels the blend never altered.

The second intervention asked whether the task becomes easier when the network only describes the difference. It learned the signed correction that turns the blend back into the target:

$$R = X - Y, \qquad \widehat{Y} = \Pi_{[0,1]}(X - f_\theta(X)). \tag{4}$$

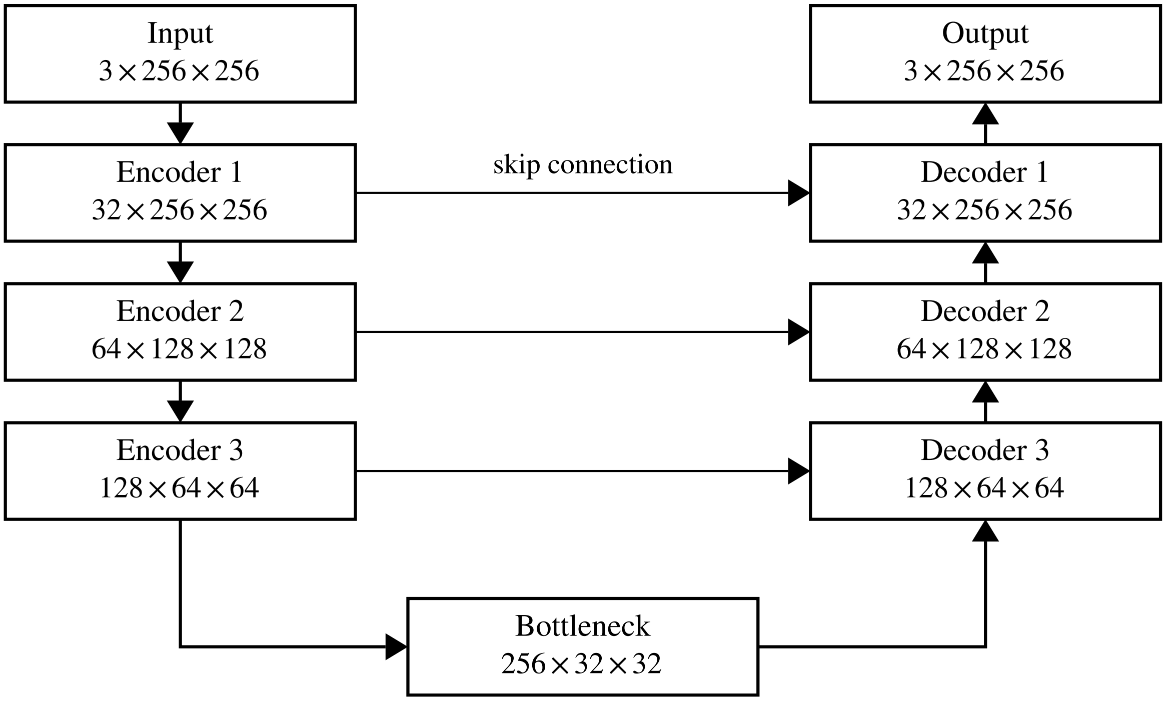

Figure 2. The fixed 256×256 U-Net used for direct reconstruction, signed correction, balanced hard-case exposure, and weighted-loss training. Each labeled block applies two convolution–normalization–activation sequences; max pooling performs each downsampling step and transposed convolution performs each upsampling step. The three skip connections concatenate encoder features into the matching decoder stage, restoring spatial detail carried by the blend rather than by isolated source truth. The network has 1,927,075 trainable parameters.

Here R is the training target and $f_\theta(X)$ the predicted correction, subtracted from the blend and clipped to recover the reconstruction. This should help because R is near zero over most of the frame, which concentrates learning on the overlap. It is important to note that the correction can be negative, since the simulator blurs the target, adds noise, and clips the blend; it is a bookkeeping quantity, not a physical contaminant layer that could be read as the companion galaxy.

The third intervention changed the training distribution rather than the model. Hard overlaps are rare under uniform sampling, so a model can score well on average while failing where it matters. Balanced sampling placed 30% of training examples in high-overlap or core-obstructed regimes and 20% in brightness/size stress, leaving 50% normal. It cannot establish that the reweighted distribution matches a real survey; it matches the stress regimes the simulator was asked to produce.

The fourth intervention changed where in the image the loss paid attention. Even with a residual target, an error averaged over all pixels lets a model trade a small gain across the background against a large error inside the overlap. The final loss therefore upweights the pixels that carry the blending, using the truth-defined affected mask A from Equation 2 and the training core K:

$$w(p) = 1 + 3A(p) + 2A(p)K(p), \qquad \mathcal{L} = \frac{\sum_{p,c} w(p)[\widehat{R}_c(p) - R_c(p)]^2}{3\sum_p w(p) + \epsilon}. \tag{5}$$

The weight $w(p)$ equals 1 on background pixels, 4 on affected pixels, and 6 on affected pixels inside the core; $\mathcal{L}$ is the weighted mean squared error over pixels p and channels c, normalized by the total weight so reweighting cannot by itself change the loss scale, and ϵ prevents division by zero. An error in the affected core therefore costs six times what it costs in the background. The limitation is structural: A and K are defined from the truth, so this loss exists only for synthetic scenes. Table 3 summarizes the progression as one design sequence rather than four separate epistemic arguments.

The weighted model trained for 20 epochs with batches of eight and Adam at 10^{-3}, on 8,000 blends with 1,000 for validation. The minimum validation loss selected epoch 20, which was also the final epoch, so the run gives no

Table 3. Reconstruction progression.

Intervention	Technical change	Observed outcome
Direct image	Predict Y with a sigmoid head	Learned the synthetic target
Signed correction	Predict $X - Y$ with a linear head	Improved hard cases
Balanced exposure	Sample difficult overlaps deliberately	Reduced rare-case failures
Weighted loss	Weight affected and core pixels	Improved grouped overlap error
Coordinate request	Add one Gaussian input plane	98.0% prompt-swap success

Table 4. Weighted-loss model on group-disjoint development suites.

Suite	Identity MSE_A	Model MSE_A	Ratio	Worse
Normal	0.0668139	0.00231890	28.81	0/1,000
Hard stress	0.0725308	0.00458983	15.80	3/1,000
Compact-bright	0.0801469	0.00872771	9.18	2/1,000
High-core	0.0778714	0.00491680	15.84	1/1,000

Note—Ratios are identity affected-region MSE divided by model MSE. The truth-defined mask selects pixels where identity already has error.

evidence that training had converged rather than simply stopped. Seeds, checkpoints, and replay commands appear in Appendix B.

4.2. *Grouped reconstruction results*

On group-disjoint normal development scenes, the unchanged blend had affected-region MSE 0.0668139 and the weighted-loss model had 0.00231890, a 28.81× ratio with 0 of 1,000 scenes worse than identity. The direction of that result is not in doubt, but its interpretation is partly circular. By Equation 2, the affected mask is exactly the set of pixels where the identity image already differs from truth by more than 0.02, so the ratio measures how well the model repairs pixels selected for being damaged. It is not a neutral comparison against a fitted astronomical baseline.

The advantage survives harder blends with smaller margins. The hard, compact-bright, and high-core suites retained 15.80×, 9.18×, and 15.84× ratios, and the model was worse than identity in 3, 2, and 1 of 1,000 scenes. Compact-bright contaminants were hardest, which follows from the clipping in Equation 1: a small, very bright contaminant drives the pixels it lands on to the display ceiling, and the information needed to undo it is no longer in the observation. These are one-seed development results.

4.3. *The coordinate request*

Reconstruction alone answers the wrong question when a scene holds two plausible sources, because the model cannot be told which one is wanted. The coordinate campaign tested whether a positional request could supply that instruction. It used a smaller U-Net for 60×60 $g/r/z$ scenes: two encoder blocks produced 16 and 32 channels, and a 64-channel bottleneck fed a bilinear decoder of GroupNorm and SiLU blocks, with a linear head returning three detected-electron bands. The request is an extra input plane holding a unit-peak Gaussian of width $\sigma_p = 2$ pixels centered on the wanted source, concatenated to the blend:

$$\widehat{Y}_q = g_\phi([X; P_q]). \tag{6}$$

Here P_q is the request plane for source q, $[X; P_q]$ denotes channel-wise concatenation, and g_ϕ is the compact U-Net with parameters ϕ. The extra plane added 144 first-layer weights, increasing the model from 118,947 to 119,091 parameters.

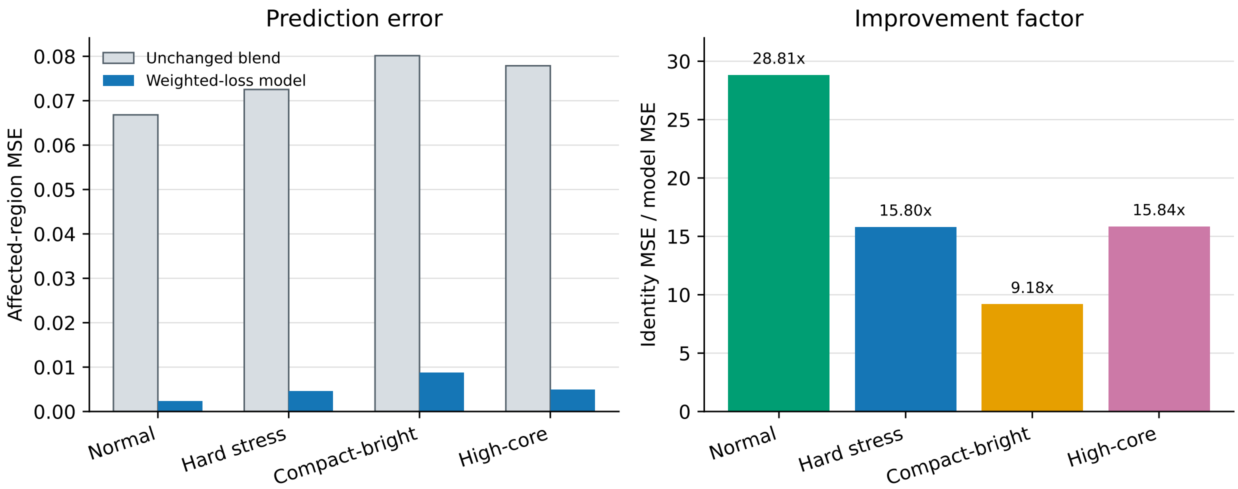

Figure 3. Group-disjoint development performance for the weighted-loss model. All four suites favor the model over the unchanged blend on the truth-defined affected region. The comparison measures synthetic correction quality, not survey photometry or recoverability.

The prompt worked as an instruction. On byte-identical randomized development scenes, changing only the coordinate changed every output, the model selected the requested source in 98.0% of swaps, and it produced effectively identical outputs in only 0.2%. Two adversarial checks behaved as predeclared: empty prompts triggered the hallucinated-source indicator in 100% of scenes and wrong prompts caused source confusion in 99%, which confirms the network reads the plane rather than ignoring it. Mean source-region MSE fell from 2.02931×10^6 without the prompt to 1.02004×10^6 with it. However, the prompted model won only 499 of 1,000 paired comparisons, so the gain in the mean comes from preventing a minority of large failures rather than from improving typical scenes. The coordinate established which source was being asked about. It did not establish which photons belonged to it.

5. WHY MODEL-SIDE CHANGES DID NOT RESOLVE RECOVERY

The favorable U-Net results did not end the architecture search. A deterministic reconstructor returns exactly one image, so if several source decompositions were equally compatible with the blend, that model could not have shown them. The next hypothesis is therefore that the alternatives existed all along and the architecture was hiding them. This section tests it six times, once for each model-side mechanism available within this project: sampling, normalizing flows, multiple output heads, separate decoders, physical output constraints, and post-hoc confidence.

Three measurement terms are needed first, because every gate below is stated in them. A model produced an *ambiguity witness* when it returned a set holding at least two decompositions that both reproduced the observation and differed scientifically by more than a fixed amount. That amount is the *primary scientific distance*: the largest discrepancy across normalized image, per-band flux, color, and centroid, so a pair counts as different only if it differs in a quantity an astronomer would report. A candidate set *covers* a truth when at least one forward-consistent member lands within distance 1.0 of it, scored against the scene's own truth, the paired observation-equivalent truth, and both at once, with a cross-decoded check evaluating a latent from one member of a pair under the other member's observation. Table 5 holds the thresholds.

The probabilistic U-Net tested whether a deterministic model was concealing valid alternatives that random sampling would expose. We drew $K = 32$ prior samples for each of 50 frozen atlas observations using one training seed. The samples varied, but produced only 24 qualifying witnesses and no coverage of either approved truth. Stochastic variation alone therefore did not recover the required alternatives: the model was uncertain about details that did not matter and confident about the allocation that did.

The conditional-flow branch asked whether a more expressive posterior would succeed where simple sampling failed. It drew $K = 32$ samples for 512 ordinary prompt evaluations and 500 near-collision evaluations. Ordinary-own, near-own, and cross-decoded alternate coverage were all zero, and the median best distances were 6.4, 7.8, and 7.8 times the acceptance threshold, so the branch was stopped and no flow was built. The margin is what matters: these were not near misses but samples far outside the region an acceptable answer would occupy.

Table 5. Declared success rules for the model-side experiments, and what each returned. Every rule was frozen before the run.

Model family	**Declared rule**	**Observed**
Probabilistic U-Net	≥ 30 witnesses; own coverage ≥ 0.70; alternate ≥ 0.30	24 witnesses; 0 coverage of either truth
Conditional flow	candidate within primary scientific distance 1.0 of truth	0 coverage; median best 6.4, 7.8, 7.8$\times$ threshold
Multiple-output model	nonzero own, alternate, and both-mode coverage	all three 0; prompt swap 0.992
Two-expert decoder	≥ 0.90 on six rates; ordinary diameter ≤ 1.0	prompt swap 0.953125; coverage 0; diameter 5.166
Physical output contract	nonnegative sources summing to the observation	14,000 of 14,000 episodes infeasible
Post-hoc calibration	a population containing both safe and unsafe outputs	0 of 7,591 eligible outputs safe

Note. Primary scientific distance is the largest normalized image, per-band flux, color, or centroid discrepancy. Sample budgets and per-branch conditions are given in the text. These are single-seed diagnostics under frozen gates, not estimates of how often each family fails in general.

The shared multiple-output model tested a simpler idea, that two output slots could hold a different decomposition in each. It emitted $K = 2$ candidates and trained one fit for 30 epochs on 19,000 ordinary and ambiguous observations. All three required coverages were zero, although set-level prompt swapping reached 0.992. The slots responded reliably to which source was requested and still never contained either approved decomposition.

The independent two-expert decoder gave the sharpest diagnostic, because it removed the most plausible remaining excuse: if one shared decoder cannot represent two decompositions, two separate decoders should. Its training-only microset held 64 observations and received one 400-epoch fit. Prompt swapping reached 0.953125 and the aggregate forward gates passed, so the model worked as a model. Every truth-coverage rate was nevertheless zero, and the ordinary expert diameter was 5.166: the experts disagreed with each other about five times more than the acceptance rule allowed, while agreeing with neither truth.

5.1. *What the loss preferred*

That result changes what needs explaining. The two-expert model had two independent branches, valid truth targets in training, and passing forward-consistency and prompt-swap gates, yet returned no truth coverage. Either the optimizer never found the truth, or the truth was never what the objective asked for.

The next test removes the optimizer from the question. For each of 64 rows we took the exact source pair and a blended compromise and scored both directly under the same training objective, without fitting anything. If the objective were well specified, the exact truth would score better on essentially every row, because it is the answer the training data was built from.

That is not what happened. On 54 of 64 rows, the loss assigned a better score to a blended compromise than to the exact source pair; the exact truth was objective-optimal on only 10 of 64. The optimizer was therefore not merely missing the truth. In most rows, the objective itself preferred another answer, so a model that minimized its training loss perfectly would still not have returned the true decomposition. Training found what the objective specified. The objective specified the wrong thing.

What this does not establish matters equally. The audit rests on one 400-epoch fit over a 64-observation microset under frozen gates, so it shows that the objective was misspecified on these rows without measuring how often losses of this form are misspecified across scenes, seeds, or survey conditions, and without identifying which term in Equation 5 is responsible.

5.2. *Four kinds of failure, and which one this was*

"The model failed" can mean four different things with four different remedies.

A *representation failure* means the network cannot express the correct answer, even if training were perfect; the remedy is a different or larger model. An *objective failure* means the answer can be represented, but the loss scores another answer more favorably; the remedy is a different loss. An *optimization failure* means the objective prefers the correct answer, but training does not reach it; the remedy is a better optimizer, schedule, or initialization. An *information failure* means several answers remain compatible with the observation itself; no change to the model can fix it, and the remedy is more data or a stronger declared assumption. Increasing network capacity directly addresses only the first case.

Read against that taxonomy, the two-expert and loss-geometry results point at the second. The network could represent strong candidate solutions, and two independent decoders were available to hold them, but the loss often scored a blended compromise below the exact truth. This is evidence of objective misspecification rather than insufficient capacity.

5.3. *Physical constraints and the calibration dead end*

Physical constraints were the next hypothesis, and they repaired a different problem than expected. The all-nonnegative, exact-sum contract required the requested source, the companion source, and the residual to be nonnegative and to sum pixel by pixel to the signed observation, which is what conservation of light would demand. It proved infeasible on the data. A target-sum exceedance occurs when the requested-plus-companion truth is already larger than the observed value at a pixel, leaving no nonnegative residual able to close the sum; noise and the finite PSF make such pixels unavoidable. Every frozen episode contained at least one, so all 14,000 of 14,000 episodes were infeasible and the contract was rejected before any model was built. Allowing only the residual to be signed passed all 23 physical gates, represented all 14,000 target pairs, and produced zero negative source values, but prompt identity across three micro fits was 0.5000 and 0.5625 on the difficult and eight-scene conditions against a 0.90 gate. Nonnegativity and signed closure repaired the bookkeeping without identifying which galaxy owned the light.

The last model-side option was to accept the outputs and learn when to trust them, but a confidence model needs examples of both safe and unsafe outputs to learn a boundary between them. After a probabilistic-model batch correction, 7,591 complete outputs were eligible under the frozen safety rules, and zero passed. The proposed post-hoc classifier therefore had no positive class. Calibration cannot create safe reconstructions that are absent from the population it is trained on.

Taken together, these six experiments changed the next question. Further architecture search could still improve an average image metric, but none had shown why one physical separation should be preferred. The project therefore turned to the forward model and the information supplied to it.

6. THE RECOVERABILITY AUDIT

6.1. *Do alternative decompositions actually exist?*

Section 5 concluded that the training objective often preferred a compromise over the truth. That leaves a prior question unanswered: if the objective were repaired, would the observation single out one answer? The ambiguity atlas tested this by constructing alternatives by hand. For a frozen scene we build a second pair of sources that sums to the same blend, then ask whether the substitute pair is scientifically different from the original.

All 50 constructed observation-equivalent source pairs reproduced the same blend, and in 19 of those 50 cases the alternative also remained inside the same model family, meaning both members were still describable by the galaxy model the fits were allowed to use. The in-family count matters most: an alternative outside the allowed family can be dismissed as arbitrary pixel noise, but an in-family alternative is an answer the method would have been willing to report.

6.2. *Why the alternatives exist*

The reason is arithmetic, and the derivation is also the reason no purely model-side fix could have worked. A telescope pixel does not record two galaxies; it records how much light arrived at that position, which is the sum of whatever each source contributed. For two sources under a noiseless additive contract,

$$y = x_1 + x_2, \tag{7}$$

where y is the observed blend and x_1 and x_2 are the two source images we would like to recover. Now suppose we take some light from one source and give exactly that much to the other. Call that transfer δ. The first source becomes $x_1 + \delta$, the second becomes $x_2 - \delta$, and their sum is

$$(x_1 + \delta) + (x_2 - \delta) = y. \tag{8}$$

The $+\delta$ and the $-\delta$ cancel, so the observation is exactly what it was before. The two galaxies have changed. The image has not. Any δ that leaves both sources feasible therefore produces a second answer that no amount of looking at y can rule out.

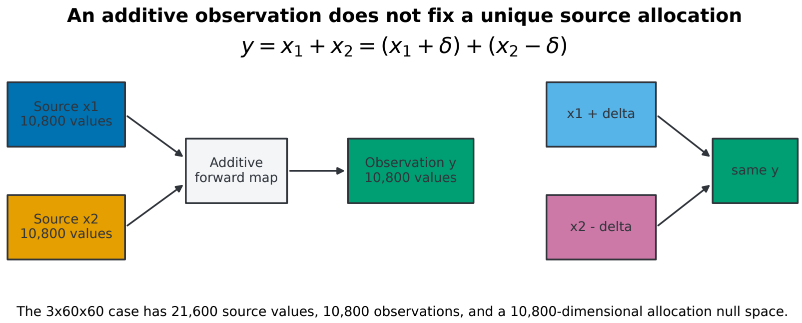

Figure 4. An additive observation records the sum, not the allocation. Transferring δ between sources leaves the blend unchanged. The diagram counts transfer directions under the unrestricted contract; it does not measure their photometric extent.

Counting these independent transfers is a linear-algebra exercise. Two $3 \times 60 \times 60$ sources hold 21,600 values between them, while the blend holds only 10,800, so the forward map on the stacked sources is $[I\ I]$, and

$$\mathrm{rank}[I\ I] = 10{,}800, \qquad \mathrm{nullity}[I\ I] = 10{,}800. \tag{9}$$

The rank is the number of independent constraints the observation supplies, so 21,600 unknown source values are constrained by 10,800 observed values, and the 10,800 missing constraints are exactly the transfer directions of Equation 8.

This distinction matters because 10,800 is an easy number to over-read. The nullity counts independent directions in which the source allocation can change without changing the blend. It does not measure how much the inferred brightness, size, or shape changes along those directions. Under nonnegativity no source value may fall below zero, so each transfer is bounded by $-x_1 \leq \delta \leq x_2$ and the admissible set is a bounded polytope rather than an unbounded subspace. Its dimension is a lower-bound description of the ambiguity's shape, not a measure of its size. The scientifically relevant quantity is its extent in flux, half-light radius, ellipticity, or whichever measurement is used downstream.

That extent is what this project cannot report: multi-start fits did find observation-equivalent solutions on the selected scenes, but the saved outputs do not preserve per-scene requested-source flux spread in reportable form. The audit demonstrates multiplicity while leaving photometric severity unquantified.

6.3. *How much does a strong assumption help?*

Real galaxies occupy a far smaller family than arbitrary nonnegative images, so the atlas result is a worst case, and the next experiment asked how much of the ambiguity a strong assumption removes. The strongest available test supplied the fit with the correct morphology family and the exact per-source brightness in each band, and seven of eight selected scenes then met the strict uniqueness rule. This is an upper bound rather than a performance figure, because the fit received information derived from the hidden sources that no real observation would supply.

Removing the hidden source brightness while keeping the shape family produced zero strict unique scenes. Six were near-unique and two were partially identifiable, so the shape restrictions did reduce the alternatives sharply; they simply did not leave one authorized answer. It is important to note that the two campaigns also differed in renderer, noise, and gates, so the contrast between seven of eight and zero of eight is not a single-variable ablation and does not show that hidden brightness alone caused the difference.

Scene 51 exposed a different boundary. Its core was more compact than the permitted shape family allowed, so no member of that family could fit it. A failed fit can therefore mean that the truth lies outside the model, not that the observation is ambiguous within it. Those outcomes look alike from the optimizer's side and mean opposite things scientifically, so recoverability statements must name both the observation and the admissible source family.

6.4. *Color as information*

The three $g/r/z$ bands are not merely 10,800 interchangeable pixel values. When a spectral energy distribution (SED) prior couples the bands, a transfer that is invisible in one band may imply an implausible color in another, which is a central feature of multiband methods such as SCARLET (P. Melchior et al. 2018). The additive count above assumes no cross-band relation, so a valid SED prior would reduce the admissible set below the bound derived here. However, the frozen records do not document whether the shape-family campaigns enforced cross-band SED coupling, leaving that reduction unmeasured. Testing explicit, misspecification-aware SED coupling is a first candidate for the expanded project.

6.5. *Does a second exposure add independent information?*

We next tested whether a second exposure of the same scene supplies genuinely independent information, or only a second look at what was already in hand. The question carries a confound, because a second exposure taken under different conditions differs from the first in two ways at once: there is more data, and the blurring pattern has changed. The campaign therefore used three conditions. S1 used one observation. S2 used two observations with the same PSF and different noise, adding data without adding blur diversity. P2 used two observations with different PSFs, adding both. Comparing P2 only with S1 would confound another exposure with a different blur pattern; S2 separates them.

P2 improved the composite local stability-and-information diagnostic in 15 of 16 scene-and-model comparisons relative to S1. The S2 control then changed the interpretation: 3 of the 16 gains were attributable to PSF diversity, 2 to the extra exposure alone, and 11 had no assignable gain under the frozen rule. Strict uniqueness under P2 remained zero of eight scenes. The 15/16 count is also softer than it looks: one of those passes compares two zero diameters, so requiring a strictly non-zero improvement gives 14 of 16. The controlled result is that decomposition, not the 15-of-16 headline, and either count is a local diagnostic rather than a count of solved scenes. The second image made the fit steadier, but it did not make the answer unique.

6.6. *External photometry and solver corrections*

The last experiment asked whether an independent measurement a survey could actually obtain would help. An external total-brightness measurement with about 5% uncertainty reduced ambiguity in Scenes 0, 5, 51, and 73, and did not help Scenes 3, 6, 18, and 81. The result is 4/8 selected discovery scenes, not a population estimate, and the scene dependence is itself the finding: external information helps when it happens to constrain a transfer direction the observation left open.

A fit that stops early looks much like a fit with no unique answer, so one further check was needed. Increasing the optimizer budget resolved Scenes 5 and 18, separating solver failure from genuine ambiguity. Separately, an exploratory bulge-to-total-ratio pattern did not survive multiple-comparison correction and depended on information unavailable before fitting, so it remains a hypothesis. All eight scenes were chosen for mechanism discovery: 0, 3, 5, 6, 18, 51, 73, and 81. Independent validation must exclude them.

7. WHAT FOLLOWS

The results above point to one structural conclusion: a deblender that can only return an image is answering a question the evidence does not always support. If some observations permit a unique source and others do not, a single output mode is the wrong interface, and the system needs three.

It should return a reconstruction when the declared evidence leaves one scientifically defensible source. It should request another measurement when that measurement is expected to distinguish the surviving alternatives, which Section 6 showed is sometimes but not always the case. It should abstain from claiming a unique report when neither condition holds. Abstention is not a failure of the model; it is an accurate statement about the observation.

Deciding among those actions requires an audit with three layers. The first runs before reconstruction and asks whether the request is answerable at all: whether the coordinate is valid, the requested object is supported by the model, the scene is not already outside the allowed family, and the request is well formed. Scene 51 is the case it exists to catch. It is promising but incomplete, and its checks have not yet been validated as an operational screen.

The second layer searches for alternatives and is the strongest completed layer in the project. It repeats fits from different initial states, clusters the surviving solutions, measures their scientific spread, checks local geometry, tests whether the survivors remain inside the stated source family, and evaluates whether an additional measurement removes them. Every recoverability result in Section 6 came from this layer.

The third layer would learn a practical safe-or-unsafe decision from completed outputs. It is blocked, for the reason given in Section 5: the available probabilistic population contains 0 safe examples among 7,591 eligible outputs, so a classifier has no positive class to learn. That is a data problem rather than a modeling choice, and a different classifier would not solve it.

The expanded project should therefore start where this one stopped, by quantifying ambiguity in scientific units. Per-scene requested-source flux spread across observation-equivalent solutions is the highest-priority missing result, because without it the audit can say that alternatives exist but not whether they matter. The next tests should add explicit SED coupling, use independent scenes that exclude the discovery set, and measure false authorization and unnecessary abstention under one frozen contract.

8. LIMITATIONS

Each result here is bounded by the setting that produced it. The reconstruction benchmark is the most controlled and therefore the least transferable. It uses clipped display-RGB composites, a simplified foreground mask, a centered target, and a controlled noise model, so its development metrics describe controlled image restoration rather than calibrated survey photometry. The corrected split removes the exact-source leakage demonstrated in the earlier protocol, but only one grouped training seed was completed and no untouched final source pool remains, so every ratio in Section 4 is a property of that single run.

The coordinate model is similarly narrow. It covers one compact U-Net family, and the failure probes in Section 4 show it has no calibrated uncertainty and no abstention behavior, so it cannot decline to answer even when the observation does not support an answer. Its lockbox remains sealed, so no held-out performance claim has been made for it at all.

The architecture diagnostics are mechanistic rather than statistical. They used single training seeds and small frozen microsets, and the two-expert diagnostic with its loss-geometry audit rests on one 400-epoch fit over 64 observations. They show clearly that objective misspecification occurred; they do not measure how often it occurs across seeds, scene populations, or loss variants.

The recoverability campaigns carry the most qualifications. They use selected synthetic scenes, exact coordinates, known PSFs, and information contracts that change between experiments. The oracle and non-oracle experiments differ in more than brightness, so their contrast is not a clean ablation. The shape family excludes at least one observed compact core, so a failed fit can indicate a misspecified family rather than an ambiguous observation. The frozen records do not document whether the shape-family campaigns enforced cross-band SED coupling. These experiments identify mechanisms; they are not estimates of real-survey prevalence.

The most consequential gap is a missing measurement rather than a missing experiment. Flux-spread severity was never recorded in reportable form, so the project can show that observation-equivalent alternatives exist without saying how different the reported galaxy would be if another were chosen. The audit establishes multiplicity but not impact.

The main conclusion survives those limits. The models learned to reconstruct, corrected difficult blends, and responded to a source request. Those achievements were real. They were not recovery certificates. Whether one galaxy can be reported depends on the observation, the allowed source model, and any independent information available to test the remaining alternatives.

APPENDIX

A. HISTORICAL ARCHITECTURE PROGRESSION

The random-row results in Table 6 document why the project changed its output target and training distribution. Exact-pixel and exact-coordinate crossings later made this split unsuitable for source-independent performance claims. The main text therefore uses only the grouped evaluation.

Table 6. Historical random-row development results.

Internal model / suite	Model MSE_A	Ratio	Worse than identity
Thayer-Direct, normal	0.004236	16.08	···
Thayer-Direct, hard	0.009390	8.04	13/1,000
Thayer-Residual, hard	0.007069	10.69	0/1,000
Thayer-BR v0.1, hard	0.004587	16.47	0/1,000

NOTE—These values use the superseded random-row development split. They document design progression only.

B. REPRODUCIBILITY CONTROLS AND DATA AVAILABILITY

The grouped implementation is `src/models.py::UNet`, `scripts/train_grouped_v02.py`, and `scripts/evaluate_grouped_portable.py`. Training used 20 epochs, batch size 8, Adam at 10^{-3}, zero configured weight decay, seed 3042, and Apple's MPS backend. The selected checkpoint is `models/grouped_br_v02_best.pth`, with SHA-256 `eea442ff21bdfbdd74815d7b292e786f187dc9a63fea73d4adde98a4b082802b`.

The archive includes `data/Galaxy10_DECals.h5`: 17,736 Galaxy10 DECaLS cutouts with SHA-256 `19aefc477c41bb7f77ff07599a6b82a038dc042f889a111b0d4d98bb755c1571`. It also contains the group-safe source split, exact blend recipes, per-scene measurements, and the selected checkpoint. Galaxy10 DECaLS is distributed under CC BY 4.0; its record and attribution appear in `DATA_LICENSE.txt`.

The coordinate implementation is `scripts/thayer_select_prompt_ablation_common.py`. Conditions A–C are centered-unprompted, randomized-unprompted, and randomized-prompted. The archive includes all three selected model states, the training and validation tensors, and the 1,000-scene development tensor with isolated targets and coordinates.

The executable notebooks are `01_project_walkthrough.ipynb`, `02_model_and_training_demo.ipynb`, and `03_recoverability_audit.ipynb`. Together they connect the raw data, model design, short training demonstration, per-scene measurements, paper figures, and selected recoverability endpoints. A code-hidden HTML export provides the same scientific walkthrough without a Python environment.

Running `bash run_all.sh` rebuilds the walkthrough and summary calculations. Adding `--full-evaluation` replays all 4,000 grouped scenes and all 1,000 coordinate scenes from the included HDF5 data and checkpoints. Recoverability counts are recalculated from the selected scene-level endpoint evidence; the archive does not repeat every upstream numerical search.

C. OPEN MEASUREMENTS

First, what is the per-scene requested-source flux spread across observation-equivalent solutions? If the saved outputs retain shape measurements, the same analysis should report half-light-radius and ellipticity spreads.

Second, how much does an explicit cross-band SED prior shrink the admissible solution set, and how sensitive is that gain to SED misspecification? The frozen records do not document whether the shape-family campaigns enforced cross-band SED coupling.

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